Training load and structure-specific load: applications for sport injury causality and data analyses

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INTRODUCTION

How should I schedule my training? How much is too much? Coaches and sports medicine clinicians commonly face such questions when considering training and injury risk. These are highly relevant inquiries, as training load is a necessary cause of sports injury.[1 2] To provide answers, our analytical approaches should align with causal frameworks. Changes in training load (eg, acute:chronic workload ratio) has been used as an interesting exposure to injury lately[3–5] and promoted as proximal in the causal chain to sports injury.[2 6] However, the aetiology behind sports injury development is multifactorial.[1] Therefore, more variables (eg, body mass, alignment, diet, strength) than training load are necessary to robustly identify 'how much is too much'.[7] Accordingly, the purpose of this editorial is to describe the differences among the concepts 'training-load', 'structure-specific load' and 'load capacity', including the varied exposures that define them.

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Definitions

Training load
Training load represents step count, throws, distance run and/or time spent practising sport. This can be used to calculate a change in training load over time (eg, acute:chronic workload ratio or week-to-week changes), which has been used as a time-varying exposure to sports injury recently.

Structure-specific cumulative load
Can be viewed as the sum of step-specific or throw-specific loads that a certain musculoskeletal structure is exposed to during a training session. Estimation of the structure-specific cumulative load per training session involves stepwise or throw-wise quantification of the load distribution and the load magnitude.

Structure-specific load capacity
Can be defined as a certain structure's ability to withstand structure-specific cumulative load.

ATHLETES AT DIFFERENT RISKS

Sports injury prevention scientists should carefully consider how best to phrase their research questions in aetiological studies.[8] For instance, the following question 'how much training load is too much among athletes with different characteristics' can be investigated under the assumption that injury risk is highest for athletes who have dramatically increased their training load. Conversely, athletes who train at a slightly increased, similar or reduced load level are likely to have a lower risk compared with the former scenario. Finally, the situation is less clear when the progression in the level of training load has been 'modest', where certain athletes might remain at a low risk, whereas athletes with different characteristics might be at a much higher risk compared with the first group (online supplementary material 1). The question remains, why is that so?

TRAINING LOAD AND STRUCTURE-SPECIFIC LOAD

Differentiating among training load, structure-specific load and load capacity may provide some answers (figure 1). In this editorial, training load is defined as, for example, step count, throws, distance run, time spent practising sport.[7] This can be used to calculate a change in training load over time (eg, acute:chronic workload ratio or week-to-week changes), which has been used as a time-varying exposure to sports injury.[3–5 9]

However, more variables than change in training load is needed to shed light in injury aetiology.[7] In biomechanical laboratories, the combination of training load (eg, steps), body mass and vertical movement (to name a few) are used to estimate structure-specific loads per step. These 'per-step' estimates can be summed to calculate a cumulative structure-specific load per session. In large-scale epidemiological studies, though impossible to measure these structure-specific cumulative loads, we may use proxy variables to better understand the structure-specific load an athlete is exposed to in a training session and their capacity to handle it.

Magnitude-related variables (eg, body weight, vertical movement) interact with training load to produce a structure-specific load by either decreasing or increasing the load magnitude. For instance, when two athletes run an identical distance, an obese athlete will have a greater cumulative structure-specific load than a normal-weight athlete (everything else being equal). Next, applied loads may be distributed differently at a structural level, depending on athletes' distribution-related variables (eg, equipment, technique, surface). For example, loads will be distributed differently between rearfoot and forefoot strikers. Together, magnitude-related and distribution-related variables interact with training load to produce the structure-specific cumulative loads that athletes are exposed to in a training session. Finally, athletes enter each training session with certain structure-specific capacities to withstand load. Like structure-specific loads, we are unable to measure structures' capacities exactly in epidemiological studies. However, proxy capacity-related variables may be included. Previous/current injuries, time between sessions, strength and diet all affect structures' capacities to withstand a given session load.

Magnitude-related, distribution-related and capacity-related variables all influence how much training load a given athlete can tolerate before sustaining injury—the point

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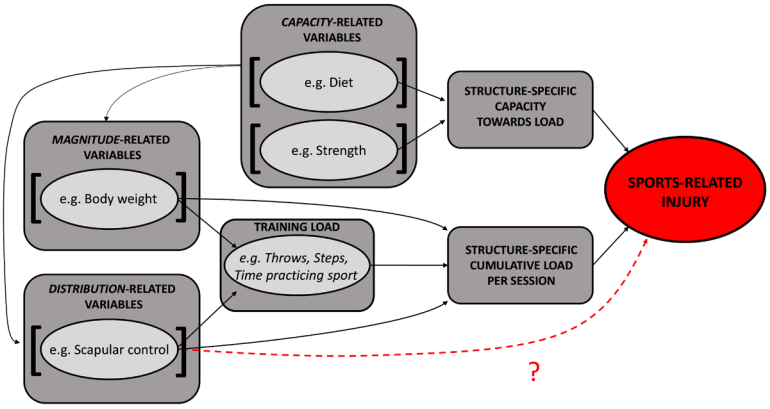

Figure 1 Simple causal diagram visualising the relationship between structure-specific cumulative load and load capacity in one training session. Variables in square shapes are unmeasurable in large-scale epidemiological studies, while variables in circles are measurable. Square brackets—[]—denote that a researcher needs to condition on these variables in an analysis with training load as primary exposure (which becomes time-varying over more training sessions, eg, acute:chronic workload ratio). Naturally, this causal diagram repeats itself over two or more training sessions.

at which the tissue-specific load exceeds the tissue-specific capacity.[7]

ANALYTICAL APPROACH

Møller and coworkers[3] examined the association between changes in handball training load and handball-related shoulder injuries across levels of distribution-related (eg, scapular control) and capacity-related variables (eg, strength). This approach was novel and transferable—informing how athlete characteristics modified the influence of training load changes on shoulder-related injury risk. This differed from traditional scientific analyses, not treating distribution-related variables as confounders, but as potential effect measure modifiers (ie, moderators).[6] Seen in figure 1, scapular control does not directly cause an injury (red arrow). Instead, it affects injury risk through an interaction with training load. Still, it is possible that one examines the association between a capacity-related (eg, strength) and a distribution-related variable (eg, scapular control) as primary exposure of interest and injury, while conditioning on training load, leaving only one path open (figure 1). However, this approach does not allow the researcher to respond to the question: 'How much is too much'. In addition, it may introduce collider stratification bias. Because of this and since training load is easy to manipulate, we encourage researchers to include training load as their main exposure of interest, while other variables are effect measure modifiers (eg, body weight, strength and scapular control).

TIME-FIXED AND TIME-DEPENDENT MODIFIERS

Typically, sports medicine researchers collect a range of potential injury risk factors, perform statistical analyses (eg, stepwise selection procedures) and observe several significant associations. However, if training load is omitted from a given analysis, then athlete subgroups at a decreased or increased risk for injury can be identified, but how and why the injury occurred remain open to informed speculation.[10] We must conceptualise how training load interacts with other variables within a training session and replace stepwise risk factor selection with analyses that reflect these conceptual mechanisms. Mechanisms that are time-varying emphasising the need to use advanced statistical methods.[11] This can be complex because certain variables, such as scapular control in the example above, were considered as time fixed within a training session. Conversely, other variables change considerably even within a training session (eg, surface, terrain, vertical movement per step).

IMPLICATIONS FOR ANALYSES

To respond to the question 'how much training load is too much among athletes with different characteristics?', researchers may need to include change in training load (eg, acute:chronic work load ratio) as a primary exposure in future data analyses, whereas magnitude-related, distribution-related and capacity-related variables that are rather time fixed may be included as effect measure modifiers if the sample size allows. If the latter variables change status within a given training session, one may need to calculate a session-specific arbitrary 'load'. Then, changes in this arbitrary load constitute the primary exposure, and capacity-related variables may be viewed as effect measure modifiers. More work is needed to define what constitutes training load in different sports (eg, time, throws, step counts, kilometres run), changes in training load (weekly ratios, acute:chronic workload or other) and how to deal with time-fixed and time-dependent modifiers in an analysis.

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